

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter Examination – 2022

Course: B. Tech.

Branch : Electrical

Semester : III

Subject Code & Name: Engineering Material Science (BTES305)

Max Marks: 60

Date:

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain concept of conductivity in metal and describe the Ohms law.	L2	6
B) Explain the phenomenon of superconductivity with application of superconductors.	L2	6
C) State and explain Meissner effect with neat diagram.	L2	6
Q.2 Solve Any Two of the following.		12
A) What is polarization? State & explain types of polarization.	L2	6
B) State and explain point defects in crystal with neat diagram.	L2	6
C) Write a short note on dielectric constant & dielectric breakdown.	L2	6
Q. 3 Solve Any Two of the following.		12
A) Draw & explain the band diagram of semiconductor. State the application of semiconductors.	L3	6
B) Explain the mechanism of conduction in semiconductors & state the properties of semiconductors.	L3	6
C) Write note on Fermi level and electron-hole concentration.	L3	6
Q.4 Solve Any Two of the following.		12
A) Classify different magnetic materials & explain their properties.	L3	6
B) Compare Soft & Hard magnetic materials.	L3	6
C) Write short note on piezoelectric, pyroelectric and ferroelectric materials.	L3	6
Q. 5 Solve Any Two of the following.		12
A) What is refractory materials? Explain properties of refractory materials.	L3	6
B) Why Galvanization & Impregnation of materials is needed for materials?	L3	6
C) Explain the X-ray diffraction using Bragg's law.	L3	6

*** End ***